Oxylipin "spoofing" by bacteria could contribute to cross-kingdom host jumps and affecting cystic fibrosis patients

Georgy Kurakin georgykurakin@gmail.com

linkedin.com/in/georgy-kurakin/ @KurakinEgor @georgykurakin.bsky.social

In a previously published preprint, we presented an analysis of publicly available data on bacterial lipoxygenases (**LOXs**), the results of which indicate their possible role in expanding the host range and even affecting representatives of different kingdoms, including **plants** and **humans** **[1]**. The association of bacterial LOXs with affecting **cystic fibrosis** patients has also been noted.

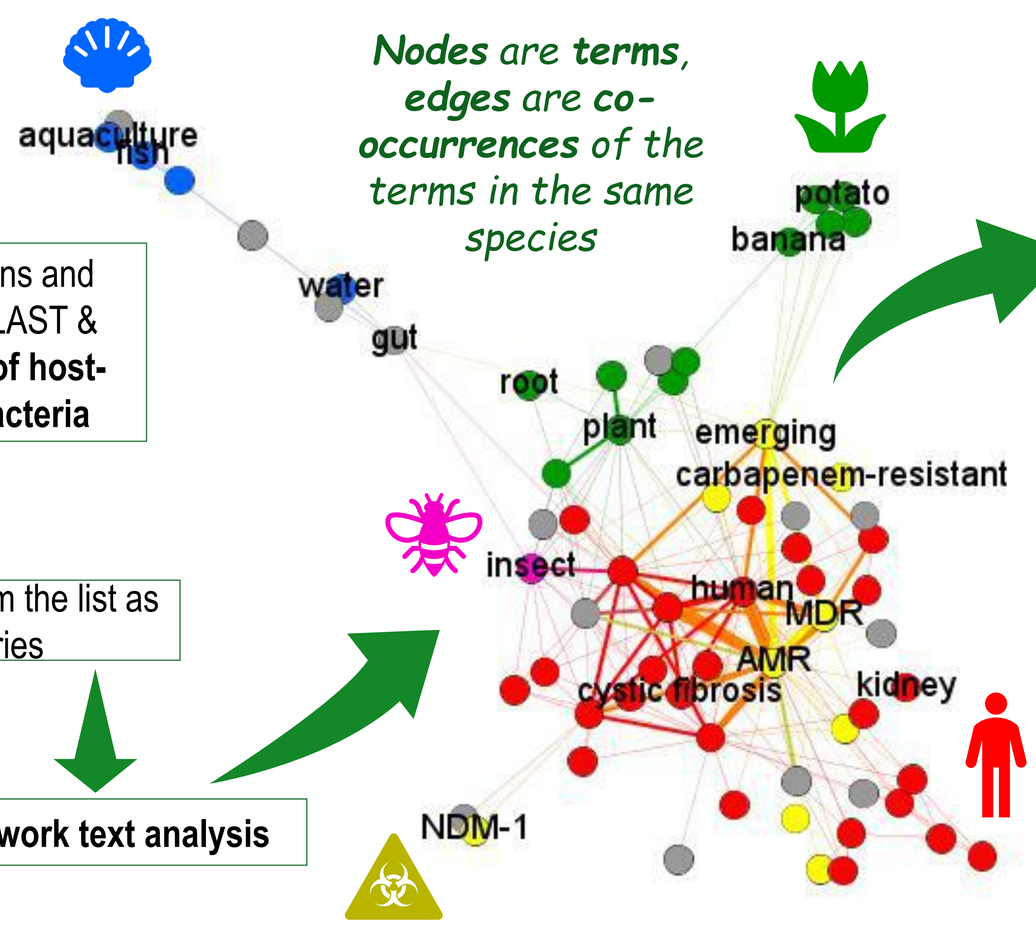
BLAST search for bacterial LOXs

Selection of LOXs of pathogens and symbionts, new iteration of BLAST & phylogenetics to obtain a **list of host-associated LOX-positive bacteria**

Using the species names from the list as **PubMed** search queries

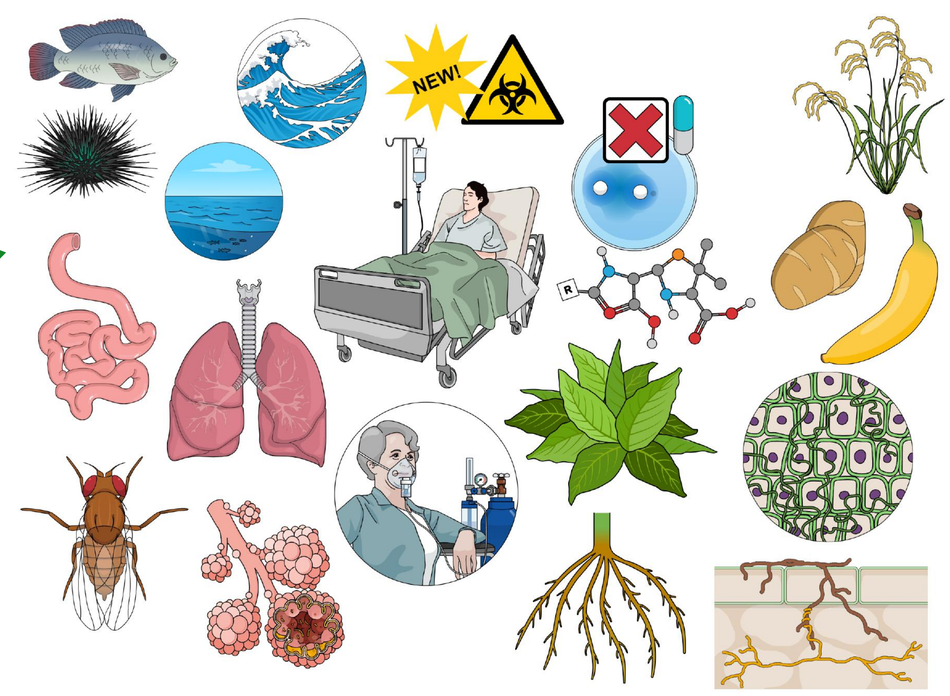
Network text analysis

Nodes are terms, edges are co-occurrences of the terms in the same species



Created with Gephi

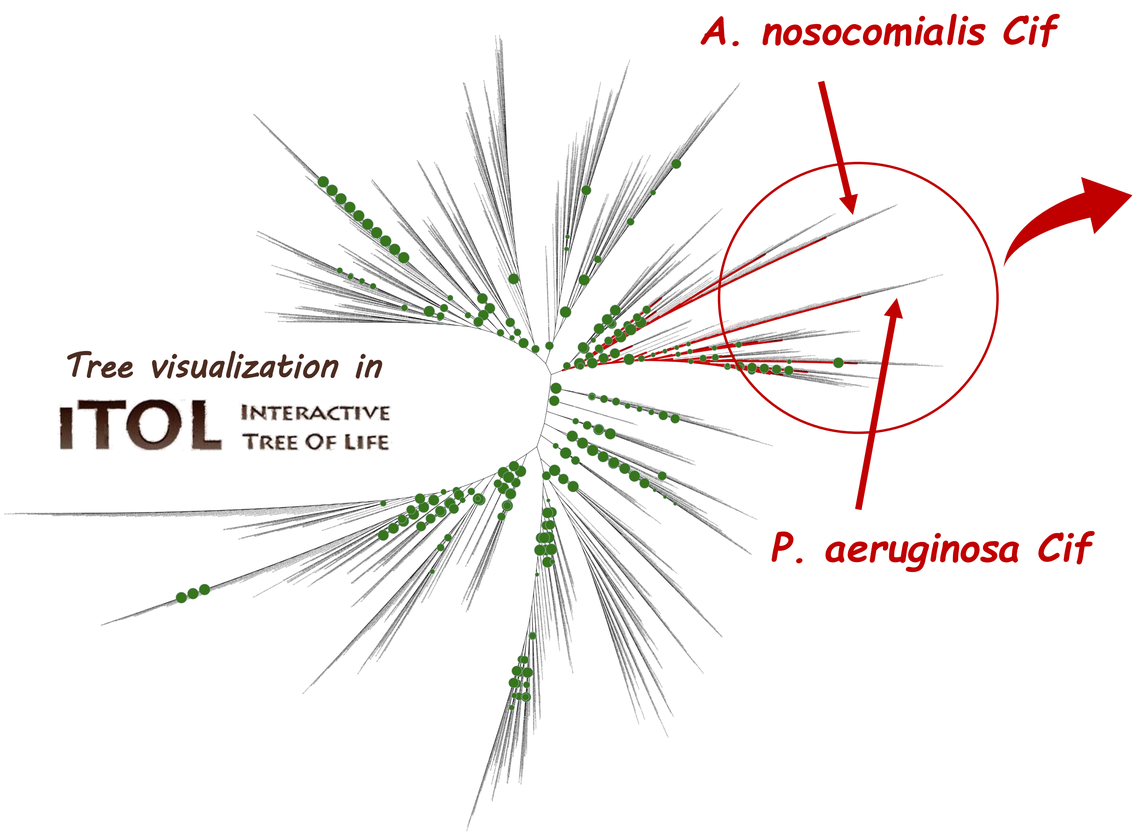
We used Mind the GRAPH

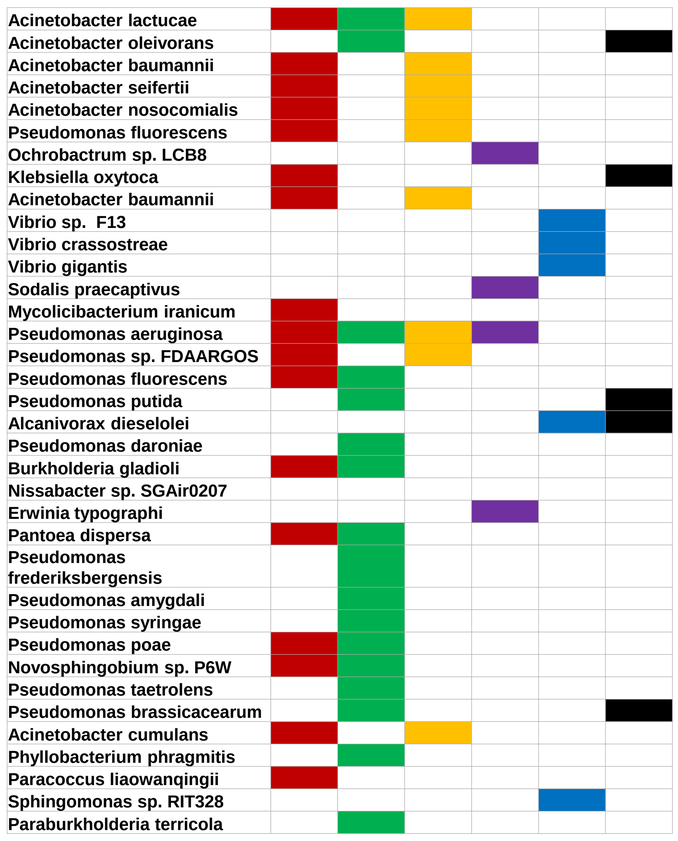


Graphic ecological "portrait" of host-associated LOX-carrying bacteria (made manually from the network)

In a literature review, we noted a similar role of **LOX** and **Cif** (CFTR inhibitory factor) of *Pseudomonas aeruginosa* in the colonization of the respiratory tract of cystic fibrosis patients **[2]**. Based on this, we decided to perform a similar analysis for Cif homologues.

We were able to identify a cluster including *P. aeruginosa* Cif and *Acinetobacter nosocomialis* Cif on a phylogenetic tree. All proteins in this cluster can be considered as **putative Cif's**. We performed a **systematic review** of the ecology of the respective bacterial species carrying these putative Cif's species from the literature in the **PubMed** database.





Host/ecological niche

- human/vertebrate
- plant
- insect
- aquatic organism
- public health concern
- biodegradation

The obtained ecological profile of bacteria carrying putative Cif's was **similar to the ecological profile** of lipoxygenase carriers discussed in the preprint **[1]**. It was similarly dominated by **human/invertebrate/nosocomial bacteria** **(29%)** and **plant-associated bacteria** **(29%)**. A high percentage of bacteria capable of **both infecting humans** and **colonizing plants** were present **(17%)**. More than a half **(52%)** of **human-affecting bacteria** had been reported as having **antimicrobial resistance**.

This **similarity of the ecological characteristics** of these bacteria might indicate **a functional relatedness between Cif's and bacterial LOXs** in the host-microbe interaction. Cif had previously been characterized as an epoxide hydrolase that cleaves polyunsaturated fatty acid epoxides and increases the pool of **immunosuppressive oxylipins [3]**.

Given this, it can be assumed that oxylipin "spoofing" is a universal strategy of bacteria to make cross-kingdom host jumps and affect patients with cystic fibrosis.

References

[1] G. Kurakin, "Bacterial lipoxygenases are associated with host-microbe interactions and may provide cross-kingdom host jumps", *bioRxiv*, 2022.06.21.497025, 2022, **doi:** 10.1101/2022.06.21.497025.

[2] A.S. Amoah, N.B. Pestov, T.V. Korneenko, I.A. Prokhorenko, G.F. Kurakin, "Lipoxygenases at the Intersection of Infection and Carcinogenesis", *Int. J. Mol. Sci.*, vol. 25, p. 3961, 2024, **doi:** 10.3390/ijms25073961.

[3] K.L. Hvorecny, C.D. Bahl, S. Kitamura, K.S.S. Lee, B.D. Hammock, C. Morisseau, D.R. Madden, "Active-Site Flexibility and Substrate Specificity in a Bacterial Virulence Factor: Crystallographic Snapshots of an Epoxide Hydrolase", *Structure*, vol. 25, pp. 697-707.e4, 2017, **doi:** 10.1016/j.str.2017.03.002.